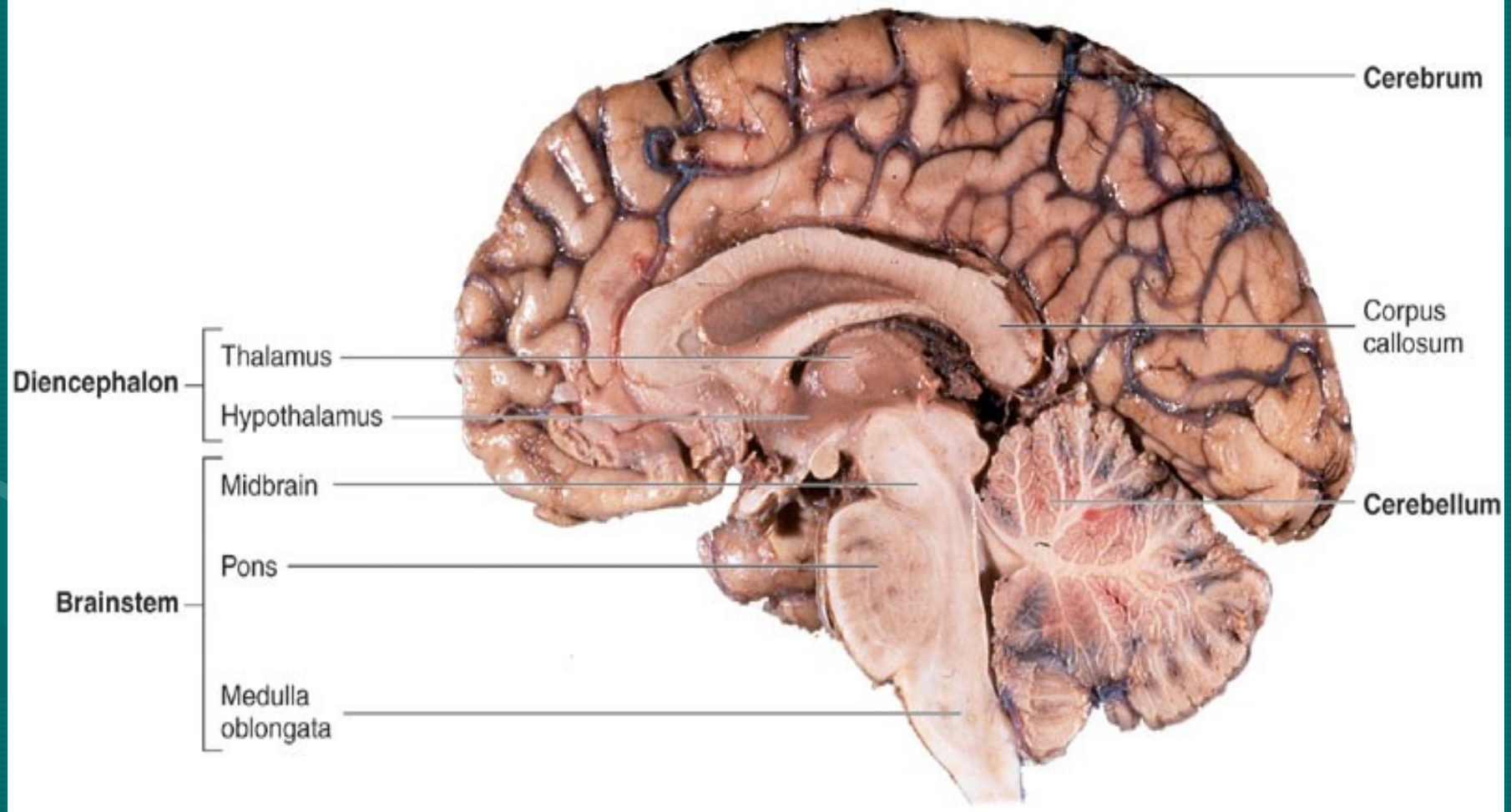
The Diencephalon

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(a)

Thalamus

Interthalamic adhesion of thalamus

Hypothalamus

Mammilliary body

Optic chiasm

Infundibulum

Pituitary gland

Diencephalon

Pons

Corpus callosum

Epithalamus

Pineal body

Cerebellum

Thalamic nuclei

Interthalamic adhesion of thalamus

(b)

Thalamus

- Located at the rostral end of the **brainstem**
- Bounded laterally by the posterior limb of the **internal capsule**
- Divided on the midline by the **third ventricle**

Divisions of the Diencephalon

Epithalamus

- Pineal organ
- Habenular nucleus and commissure
- Posterior commissure
- Stria medullaris thalami

Hypothalamus

- Ventral portion of diencephalon
- Largely concerned with autonomic, vegetative functions and interrelation with the pituitary gland

Subthalamus

- Between the hypothalamus and the dorsal thalamus
- Includes the **zona incerta** and **subthalamic nucleus**
- Concerned with *extrapyramidal and basal ganglia activity*

Thalamus

- “Dorsal thalamus”
- Largest and most dorsal part of the diencephalon

Metathalamus

- **Medial** and **Lateral geniculate bodies**

- The **dorsal thalamus** comprises numerous nuclear masses which fall generally into three principle categories:

  1. Cortical integration nuclei concerned with complex cerebral cortical activities

  2. Relay nuclei involved in projecting information from other regions of the nervous system towards various primary cortical regions

  3. A group of nuclei concerned with vegetative and autonomic activity relaying to the functions of the hypothalamus and brainstem reticular formation

- On coronal section, a sheet of white matter (internal medullary lamina) may be seen in the middle of the thalamic grey matter
- Thalamic nuclei may also be grouped according to their relationship to this lamina;

  a. **dorso-medial group** and

  b. **ventro-lateral group**

The Ventro-Lateral Nuclei

The group of nuclei falling into the ventro-lateral group may be conveniently divided into a ***ventral tier*** *and a* ***dorsal tier***

Ventral Tier Nuclei

- **Anterior Ventral** and **Lateral Ventral**

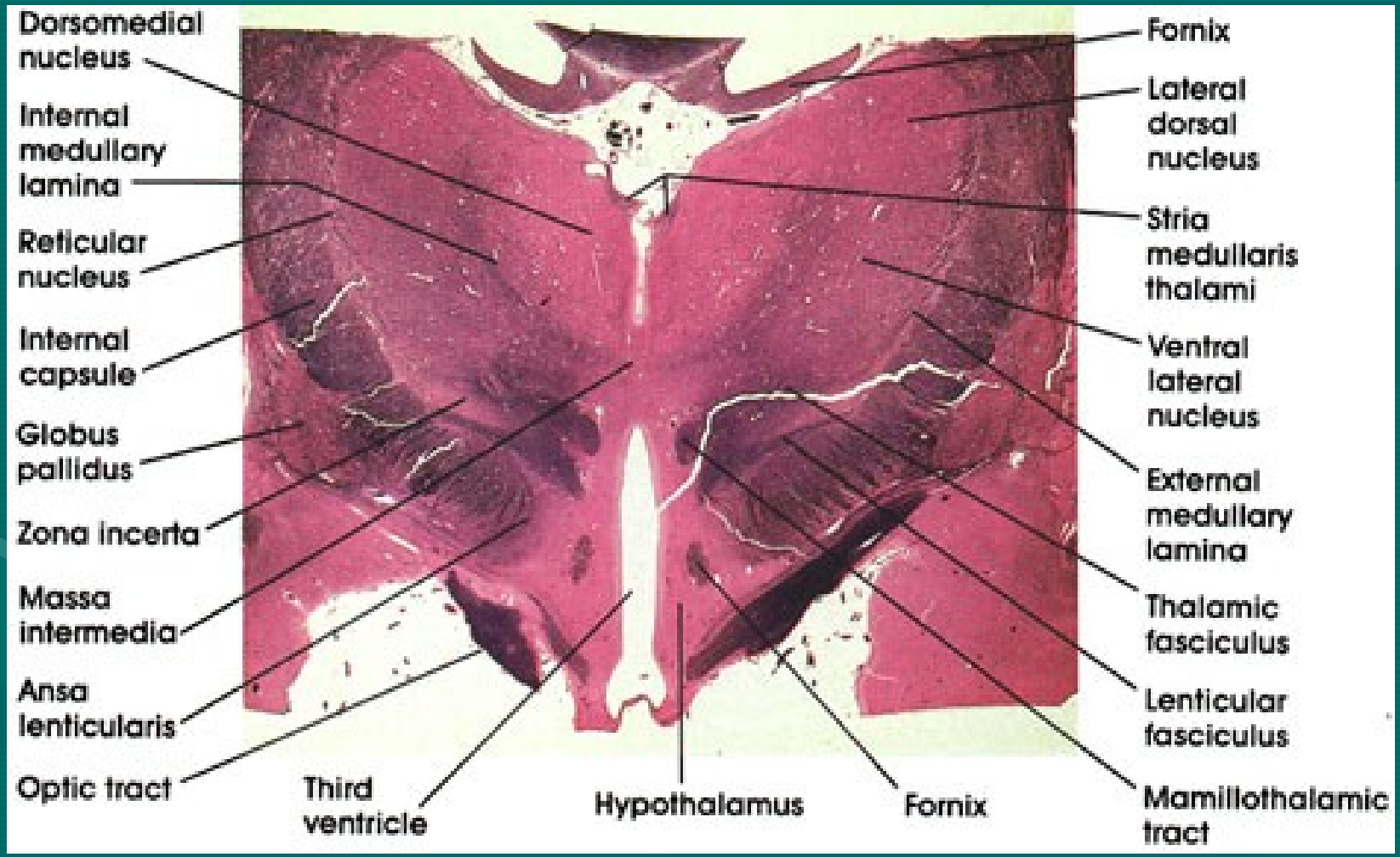
  - related to the *basal ganglia* and *cerebellum*
- **Posterior Ventral (medial), Posterior Ventral (lateral), Medial geniculate body, Lateral geniculate body**

  - specific sensory relay nuclei

Ventral anterior nucleus coronal



Ventral lateral nucleus coronal



Dorsal Tier Nuclei

- **Pulvinar** and **Lateral**

  - Cortical integration nuclei, related to *association areas* of cerebral cortex

Principle Neural Connections of the Thalamic Nuclei

- Ventral Tier -

ANTERIOR VENTRAL

- Afferents – Thalamic fasciculus from basal ganglia (Receives small input from the substantia nigra
- Efferents – to motor and premotor cortex (also to some basal ganglia)

LATERAL VENTRAL

- Afferents – brachium conjunctivum from cerebellum
- Efferents – to motor and premotor cortex

POSTERIOR VENTRAL (lateral division)

- Afferents – medial lemniscus and spinal lemniscus (general sensation from the body below the head)
- Efferents – to the post central gyrus (general sensory cortex)

POSTERIOR VENTRAL (medial division)

- Afferents – trigeminal lemniscus (general sensation from the head and the face)
- Efferents – to the inferior portion of post central gyrus

- The 2 divisions of the posterior ventral nucleus together constitute the ***largest sensory relay nucleus of the thalamus***

MEDIAL GENICULATE BODY

- Afferents – brachium of the inferior colliculus
- Efferents – auditory radiations (sublenticular) to the transverse temporal gyri of Heschl

LATERAL GENICULATE BODY

- Afferents – optic tract
- Efferents – visual radiations (retrolenticular) to the primary visual cortex of the occipital lobe (calcarine cortex = cuneus and lingual cortex)

Dorsal Tier

PULVINAR

- Reciprocal (both afferent and efferent) connections with the association cortex of the ***posterior parietal, lateral occipital*** *and* ***posterior temporal lobes***

LATERAL

- Reciprocal connections with the ***parietal lobe***

Connections of the Nuclei lying medial to the Internal Medullary Lamina

DORSO MEDIAL NUCLEUS

- Afferents – from the hypothalamus and from the prefrontal lobe cortex
- Efferents – to cortex of the prefrontal lobe

MIDLINE NUCLEUS

- Consists of a thin layer grey matter adjacent to the wall of the third ventricle (including the massa intermedia – if present)
- It is concerned with the vegetative functions, and possesses reciprocal connections with the hypothalamus

- Approximately 70% of brains possess a massa intermedia

- The Internal Medullary lamina posteriorly encircles the ***centro median nucleus***, and anteriorly, the ***anterior nucleus***

ANTERIOR

- Afferents – mamillo thalamic tract from the mamillary body of the hypothalamus
- Efferents – to the gyrus cinguli of the limbic lobe

CENTRO MEDIAN

- Abundant connections with most of the other nuclei of the dorsal thalamus; also connections with the basal ganglia
- Has a role in general integration of thalamic activity

INTRALAMINAR NUCLEI

- Small scattered clusters of cells found in the substance of the ***internal medullary lamina,*** which are related to most other thalamic areas
- The brainstem reticular formation projects to these nuclei, and there is also some input derived from the spinothalamic tracts

RETICULAR NUCLEI

- Small patches of grey matter found enmeshed in the fibers of the ***external medullary lamina,*** possessing diffuse connections with the brainstem reticular formation
- Efferent projections are to most areas of the cerebral cortex
- Functionally, these nuclei are related to the reticular activating system

Mamillary body coronal



Hypothalamus

- Comprises the inferior portion of the diencephalon, and possesses numerous nuclei which may be grouped into three hypothalamic regions
- The three regions are arranged rostro caudally as follows:
  1. Supra optic area
  2. Tuberal area
  3. Mamillary area

- The hypothalamus may be divided into medial and lateral portions by the fibers of the fornix which traverse the hypothalamus *en route* to their termination in the mamillary bodies

Supraoptic Area

Nuclei:

1. Paraventricular nucleus
2. Supraoptic nucleus
3. Anterior nucleus

Paraventricular and Supraoptic nuclei are well defined nuclei whose neurons contain cytoplasmic inclusions of neurosecretory material

Tuberal Area

Nuclei:

1. Ventro medial nucleus
2. Dorso medial nucleus
3. Lateral nucleus

Ventro medial and Dorso medial nuclei are medial to the fornix

Lateral nuclei – lateral to the fornix

Mamillary Area

Nuclei:

1. Posterior Nucleus
2. Nucleus of the mamillary body

Main Afferent Connections of the Hypothalamus

Medial Forebrain Bundle

- An affferent and efferent system connecting ***most hypothalamic nuclei*** with the ***septal area*** rostrally, and the ***tegmentum of the midbrain*** caudally

Fornix

- From the hippocampus in the temporal horn of the lateral ventricle
- Precommissural termination is in the ***Supraoptic area (and septal region)***
- Post commissural termination in the ***mamillary body nucleus***
- fornix divides at the anterior commissure

Pallido Hypothalamic Fibers

- From the Globus Pallidus

Stria Terminalis

- From the medial portion of the ***amygdaloid nucleus in the temporal lobe to the anterior hypothalamus*** (and septal region)
- Courses along the lateral ventricle adjacent to the medial border of the caudate nucleus

Visceral and Somatic Sensory

- From the lemniscal system and the reticular formation

Periventricular Fibers

- From the ***midline nuclei*** and the ***dorso medial nucleus of the dorsal thalamus*** to the dorsal portion of the hypothalamus

Main Efferent Pathways

1. Medial Forebrain Bundle
2. Mammilothalamic tract

   - from the ***mamillary bodies*** to the ***anterior nucleus of the dorsal thalamus***

3. Supraoptic Hypophyseal Tract

- from the ***supraoptic and paraventricular nuclei,*** down the stalk of the pituitary to the neural lobe of the pituitary (neurosecretory, transmitting antidiuretic hormone and oxytocin)

4. Dorsal longitudinal fasciculus (of Schutz)

- travels in the ventral portion of the periaqueductal grey matter, to the ***brainstem reticular formation and autonomic nuclei***

5. Tubero Hypophyseal Fibers

- from the ***median eminence of the tuberal region*** to the vascular ***hypohyseal portal system in the pituitary stalk***

- Conveying hypothalamic “releasing factors” to the anterior pituitary, regulating release of anterior pituitary hormones

- The infundibulum (ventral portion of the hypothalamus where the stalk of the pituitary gland is attached) contains a high concentrate of acetyl choline, and also some dopamine
- Dopamine content is increased during pregnancy and lactation

Hypothalamic Functions

- The anterior hypothalamus regulates *parasympathetic responses*
- The posterior portion regulates *sympathetic activity*

Temperature Regulation

- The anterior portion of the hypothalamus respond to changes in temperature of the blood which supplies it
- **Anterior Hypothalamus** – stimulation activates mechanism resulting in heat loss, sweating and vasodilation
- **Posterior Hypothalamus** – stimulation activates sympathetic mechanisms resulting in heat conservation such as peripheral vasoconstriction
- Somatic mechanisms are stimulated resulting in somatic muscle activity (shivering)

Water Balance Regulation

- Some anterior hypothalamic neurons are sensitive to osmotic changes in blood plasma, influencing the release of the **anti diuretic hormone**

- Diabetes insipidus may be evident following damage to the **supraoptic region of the pituitary stalk**

Feeding Regulation

- Ventro Median Nucleus – satiation

  *> lesions produce **hyperphagia** = excessive eating*

- Lateral Nucleus – feeding

  *> lesions produce **hypophagia** = reduced eating*

Sleep-Wakefulness Regulation

- Mediated via connections with the ascending **Reticular Activating System**

- Lesions in the **dorso lateral mamillary region** may produce *somnolence*

Hormonal Regulation

- Release of **pituitary hormones** is regulated via releasing factors produced in the **tuberal hypothalamic area** and delivered via the hypophyseal portal vessels to the **anterior pituitary**

Regulation of Emotions

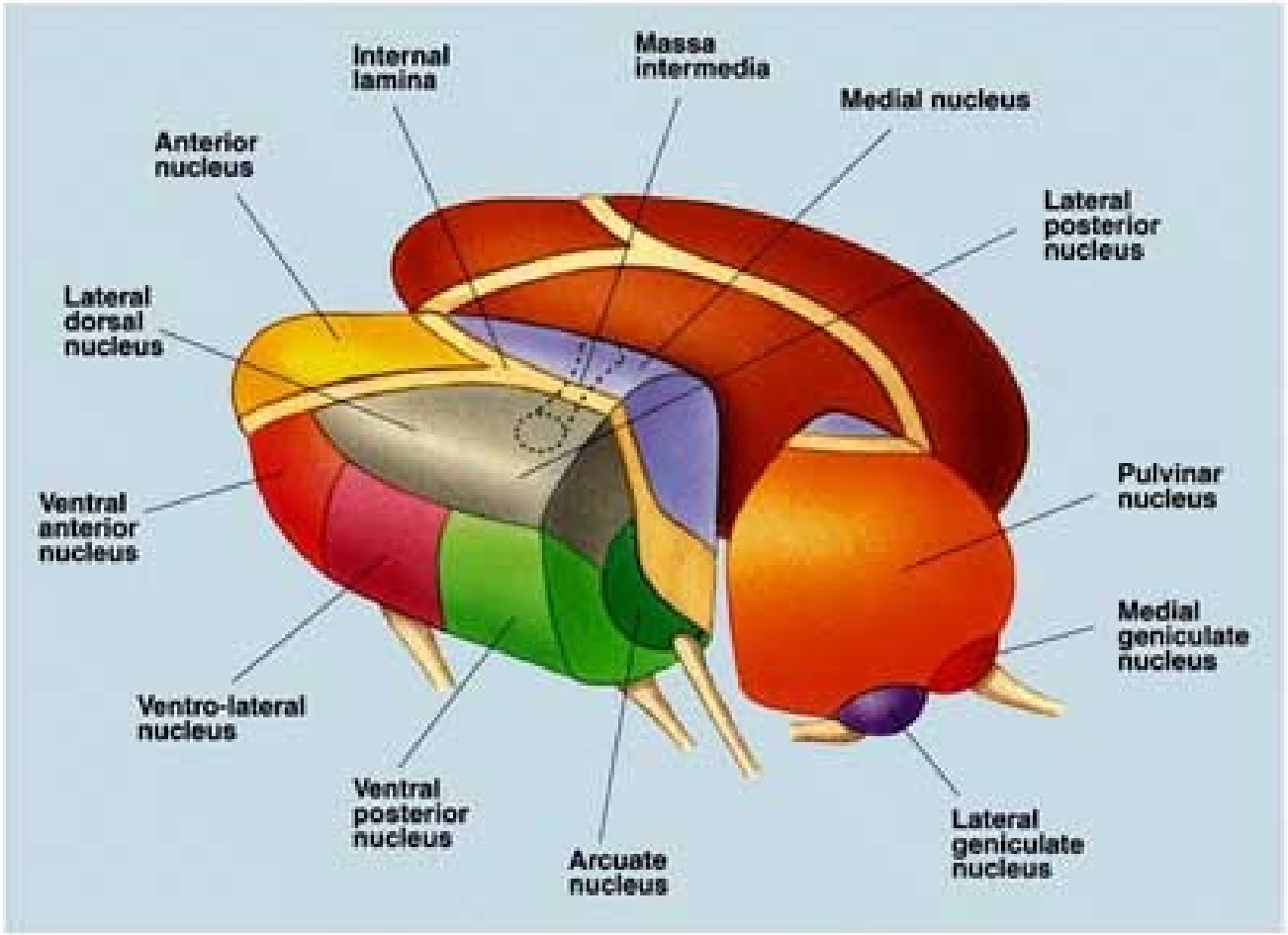
- Mediated by numerous ***interconnections with the limbic system,*** involving hypothalamic integration of limbic system effector mechanisms

- Lesions in the **ventro medial nucleus** may produce rage responses or hyper -irritability

- Sexual development may be altered in the presence of tumors in the region of the hypothalamus

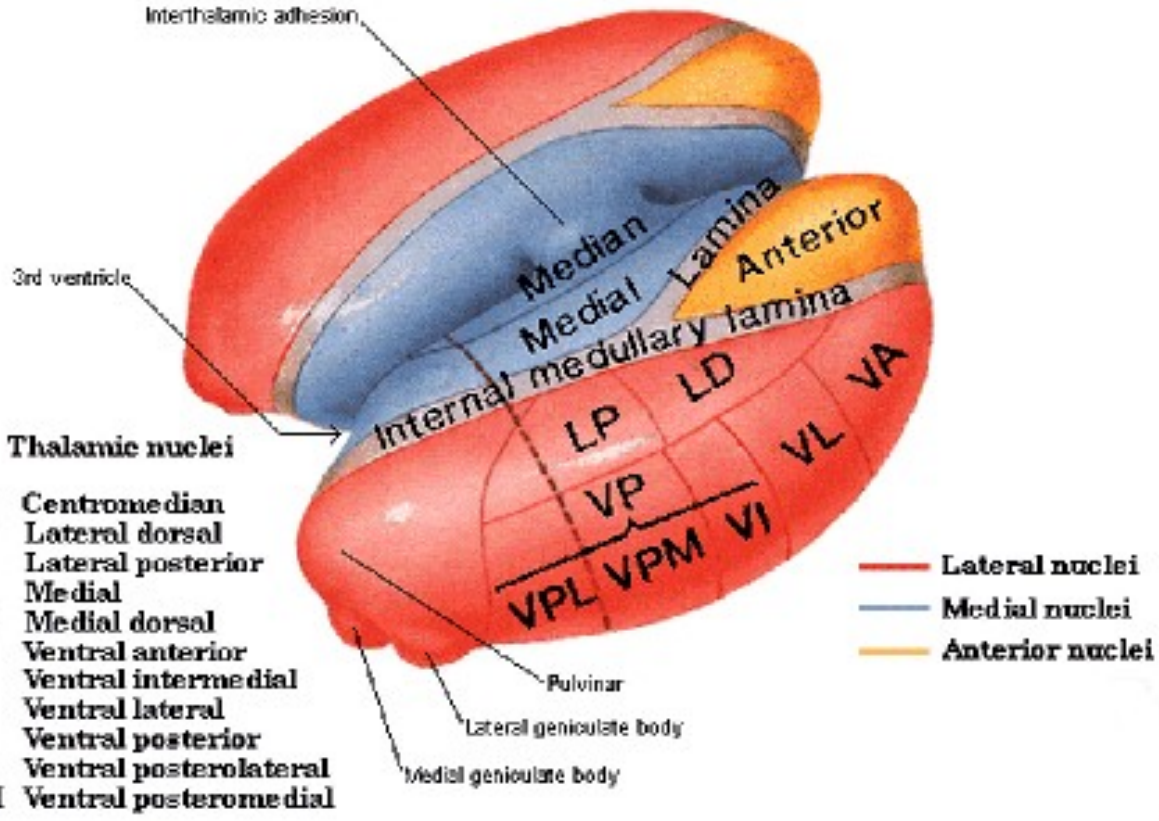
- May lead to either **precocious puberty**
- Or to **Hypogonadism** and ***failure of development of secondary sexual characteristics***

- Nuclei of the Thalamus



Thalamus

Schematic Representation of Thalamus



Thalamic nuclei

CM	Centromedian
LD	Lateral dorsal
LP	Lateral posterior
M	Medial
MD	Medial dorsal
VA	Ventral anterior
VI	Ventral intermedial
VL	Ventral lateral
VP	Ventral posterior
VPL	Ventral posterolateral
VPM	Ventral posteromedial